Sai Sumedh Kaveti

Software Engineer | New Grad, May 2027 | Raleigh, NC | Open to Relocation | saisumedhkaveti.work
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Education

North Carolina State University Raleigh, NC
Master of Science in Computer Science; GPA: 3.72/4.0 Expected May 2027
Relevant Coursework: Design and Analysis of Algorithms, Software Engineering, Computer and Network Security, DevOps, Computer Networks, Automated Learning and Data Analysis

VNR Vignana Jyothi Institute of Engineering and Technology Telangana, India
Bachelor of Technology in Computer Science (Data Science); GPA: 8.93/10.0 Dec 2021 – May 2025

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML/CSS
Backend: Spring Boot, Spring Security, JPA/Hibernate, Node.js, Express.js, REST APIs, JWT, RBAC, BullMQ
Frontend and Databases: React, PostgreSQL, Redis
DevOps: Docker, Docker Compose, Nginx, GitHub Actions, CI/CD, Git
Testing: JUnit 5, MockMvc, Vitest, Supertest, Playwright, k6, Integration Testing
Concepts: Concurrency Control, Caching, Idempotency, Asynchronous Processing, Transactions, Server-Sent Events
AI/ML: OpenAI API, Pandas, NumPy, scikit-learn

Projects

TaskFlow – Team Task Management Platform GitHub | Live Demo
Java 21, Spring Boot, Spring Security, JPA/Hibernate, PostgreSQL, React, Docker, GitHub Actions

- Migrated a Node.js/Express.js API to Java 21 and Spring Boot with no breaking API or schema changes, keeping the original backend as a rollback option.
- Implemented JWT authentication and role-based access control (RBAC), checking project membership on every request, so removed members lose access on their next request and their assignments clear in one transaction.
- Prevented lost updates using PostgreSQL row-level locks and transactional services, validated by **31 JUnit 5/MockMvc integration tests** against real PostgreSQL in GitHub Actions CI.

FinSight – Personal Finance Platform GitHub | Live Demo
TypeScript, Node.js, Express.js, PostgreSQL, Redis, BullMQ, React, Docker

- Reduced dashboard p95 latency by **83% (18.9 ms to 3.1 ms)** with Redis caching; local k6 tests used 10 concurrent users and 10,000 transactions, with throughput rising from 763 to 5,236 requests/sec.
- Built retry-safe CSV imports for up to 10,000 rows with BullMQ, idempotency keys, atomic commits, and a database outbox for recovery after queue failures.
- Kept cached reads consistent with per-user revisions updated by PostgreSQL triggers in each write transaction; validated caching and imports with **19 PostgreSQL/Redis integration tests**.

InsightChat AI – Grounded LLM Data Analysis GitHub | Live Demo
TypeScript, React, Node.js, Express.js, PostgreSQL, OpenAI API, Server-Sent Events, Playwright

- Reduced **estimated evidence-context tokens by 80–94%** on a 10,000-row dataset through question-specific column and aggregate selection.
- Computed CSV statistics in application code and supplied selected evidence to OpenAI for explanations, instructing the model to decline questions unsupported by the data.
- Streamed answers over Server-Sent Events, cancelling model calls when users disconnect and saving only completed replies; validated by **36 backend tests and 2 Playwright end-to-end tests** in CI.

Experience

Graduate Teaching Assistant (Business Analytics) Aug 2026 – Present
North Carolina State University, Poole College of Management, Raleigh, NC

- Support **162 students across 3 sections of BUS 351 and BUS 458** through 12–15 hours of weekly assistance with Python data preparation, SQL queries and joins, and analytics assignments.
- Review data analysis and predictive-modeling assignments, troubleshoot Python and SQL errors, and explain regression, classification, and model-evaluation concepts with consistent technical feedback.

Machine Learning Intern Oct 2024 – Apr 2025
Indian National Centre for Ocean Information Services (INCOIS), Telangana, India

- Built an end-to-end Python regression pipeline with Pandas, NumPy, and scikit-learn to preprocess and analyze **100+ years of global meteorological data** for climate forecasting.
- Automated preprocessing workflows and refactored temperature-data processing into reusable Python modules.